Covariant Biquaternionic Tetrads in UBT: Metric Rank, Connection Reconstruction, and Integrability

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Abstract

We formulate the local classical geometry of Unified Biquaternion Theory (UBT) directly from the covariant first derivative of its single fundamental biquaternionic field. Defining $E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta$, the classical Lorentz sector is the real four-dimensional slice $E_\mu = i e_\mu{}^0 \mathbf{1} + e_\mu{}^k \mathbf{e}_k$. Quaternion conjugation gives the exact central identity

$$\frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu) = g_{\mu\nu}\mathbf{1},$$

so the metric is not defined by a trace, real-part map, phase projection, or imaginary-time fiber average. The tetrad-to-metric map has rank ten at every nondegenerate tetrad and a six-dimensional kernel equal to local Lorentz frame freedom. We then reconstruct the connection. For every tetrad and specified torsion, the metric-compatible Lorentz connection is uniquely $\omega = \mathring{\omega}(e) + K(T)$; the torsion-free GR branch is the special case $K = 0$. Hence the remaining full-UBT question is the dynamics of torsion, not a kinematically arbitrary Ω_μ. We give an explicit affine field Θ_{SR} generating the Minkowski tetrad and solving the standard flat source-free second-order master operator. Finally, we prove that a naive one-sided connection with invertible Θ has a flatness obstruction under the stated torsion-free compatibility assumptions, while a natural two-sided biquaternionic derivative obeys $[D_\mu, D_\nu]\Theta = F^A_{\mu\nu}\Theta - \Theta F^B_{\mu\nu}$ and permits nonzero conjugate left/right curvatures. Curved-space existence, torsion dynamics, Lorentz-slice preservation by the full equations, and Einstein dynamics from the canonical UBT action remain open.

Contents

1 Scope and proof status

The purpose of this paper is to establish a minimal local geometric structure consistent with the original biquaternionic intuition of UBT and to identify exactly what remains to be proved. We do not claim an unconditional derivation of the Einstein equations.

Proved in this draft

- a projection-free Lorentzian metric from a central biquaternionic anticommutator;
- local representation of every Lorentz metric and rank ten of the tetrad-to-metric differential, with six Lorentz gauge directions;
- unique reconstruction of every metric-compatible frame connection from the tetrad and specified torsion;
- the torsion-free Levi–Civita spin connection as the unique classical GR connection;
- an explicit affine Θ representer of every constant Lorentz tetrad, including Minkowski spacetime;
- a no-go theorem for the naive one-sided invertible curved branch and the exact curvature identity for a two-sided biquaternionic derivative.

Open

- deriving vacuum torsion or spin-sourced torsion from the canonical UBT action;
- fixing the precise paired left/right connection, involution, and normalization acting on Θ;
- proving local and global existence for the implicit curved-space system $D_\mu\Theta = \sqrt{\mathcal{N}_0}E_\mu$;
- proving that the complete dynamics preserves the real Lorentz slice;
- deriving Einstein dynamics and the perturbation bridge from the original UBT master dynamics;
- canonical on-shell generation of Schwarzschild, Kerr, FRW, and wave sectors.

2 Relation to standard tetrad geometry and earlier biquaternionic work

The tetrad postulate, Levi–Civita spin connection, torsion, and contorsion used below are standard Cartan/tetrad geometry [1, 2]; they are not claimed as new theorems of UBT. The UBT-specific content is the identification $E_\mu = \mathcal{N}_0^{-1/2} D_\mu\Theta$, the central biquaternionic anticommutator metric on the Lorentz slice, its rank interpretation, the affine single-field Minkowski representer, and the conditional one-sided obstruction/two-sided intertwiner analysis. Related complex and biquaternionic gravity programmes include Refs. [3, 4]; the present paper makes no claim that standard frame geometry originates here.

3 Biquaternionic algebra and the Lorentz slice

Let $\mathbb{B} = \mathbb{C} \otimes \mathbb{H}$, with commuting complex unit i and quaternion units

$$\mathbf{e}_j\mathbf{e}_k = -\delta_{jk}\mathbf{1} + \epsilon_{jk\ell}\mathbf{e}_\ell.$$

Quaternion conjugation $\sharp$ reverses the quaternion units without conjugating the commuting complex coefficients:

$$(z^0\mathbf{1} + z^k\mathbf{e}_k)^\sharp = z^0\mathbf{1} - z^k\mathbf{e}_k.$$

Definition 3.1 (Classical Lorentz module)**.**

$$W_L = \{ix^0\mathbf{1} + x^k\mathbf{e}_k \mid x^a \in \mathbb{R}\}.$$

For $X = ix^0\mathbf{1} + x^k\mathbf{e}_k$ *and* $Y = iy^0\mathbf{1} + y^k\mathbf{e}_k$*, let*

$$\eta(X, Y) = -x^0y^0 + x^ky^k.$$

Theorem 3.2 (Central anticommutator identity)**.** *For all* $X, Y \in W_L$*,*

$$\boxed{\frac{1}{2}(X^\sharp Y + Y^\sharp X) = \eta(X, Y)\mathbf{1}.} \tag{1}$$

Proof. In scalar–vector notation, quaternion multiplication is $(a, \mathbf{u})(b, \mathbf{v}) = (ab - \mathbf{u}\cdot\mathbf{v}, a\mathbf{v} + b\mathbf{u} + \mathbf{u}\times\mathbf{v})$. For $X = (ix^0, \mathbf{x})$, $X^\sharp = (ix^0, -\mathbf{x})$. The scalar part of $X^\sharp Y$ is $-x^0y^0 + \mathbf{x}\cdot\mathbf{y}$. The vector parts of $X^\sharp Y$ and $Y^\sharp X$ are opposites, including the cross products, so they cancel in the half-sum. □

4 The covariant UBT tetrad

The fundamental field remains

$$\Theta(q, \tau) \in \mathbb{B}, \qquad \tau = t + i\psi.$$

On physical spacetime it is assumed smooth. Define

$$\boxed{E_\mu := \mathcal{N}_0^{-1/2} D_\mu \Theta,} \tag{2}$$

where $\mathcal{N}_0 > 0$ is fixed globally and D_μ is a covariant derivative. The connection is kept representation-independent at this stage:

$$D_\mu\Theta = \partial_\mu\Theta + \rho_*(\Omega_\mu)\Theta. \tag{3}$$

The map ρ_* must be specified in each concrete representation; Eq. (3) does not assume left, right, or two-sided action prematurely.

Definition 4.1 (Admissible classical tetrad)**.** *A configuration belongs to the classical tetrad sector if*

$$E_\mu = i{e_\mu}^0\mathbf{1} + {e_\mu}^k\mathbf{e}_k \in W_L$$

for all μ and the matrix ${e_\mu}^a$ is invertible.

5 Projection-Free Metric and Algebraic Bivector

By Theorem 1,

$$\boxed{\frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu) = g_{\mu\nu}\mathbf{1},} \tag{4}$$

where

$$g_{\mu\nu} = {e_\mu}^a {e_\nu}^b \eta_{ab}, \qquad \eta_{ab} = \mathrm{diag}(-1, 1, 1, 1). \tag{5}$$

The complete left-hand side is already a central real element. Hence Eq. (4) defines the metric without discarding or projecting noncentral components. The fixed constant $\mathcal{N}_0$ only sets units.

The complementary antisymmetric product is

$$\Sigma_{\mu\nu} := \frac{1}{2}(E_\mu^\sharp E_\nu - E_\nu^\sharp E_\mu), \qquad \Sigma_{\nu\mu} = -\Sigma_{\mu\nu}. \tag{6}$$

Thus one product contains both the symmetric central geometry and an oriented bivector sector. No gauge-field identification of $\Sigma_{\mu\nu}$ is made without a separate transformation and dynamical analysis.

6 The local rank theorem

Define $\Phi(e) = e\eta e^{\mathsf{T}}$.

Theorem 6.1 (Full local metric rank)**.** *At every invertible tetrad, the differential $d\Phi_e : \mathbb{R}^{4\times 4} \to \mathrm{Sym}_4(\mathbb{R})$ is surjective and has rank ten. Its kernel is the six-dimensional local Lorentz algebra.*

Proof. The first variation is

$$\delta g_{\mu\nu} = \eta_{ab}(\delta e_\mu{}^a e_\nu{}^b + e_\mu{}^a \delta e_\nu{}^b).$$

For an arbitrary symmetric $h_{\mu\nu}$ choose

$$\delta e_\mu{}^a = \frac{1}{2} h_{\mu\rho} e^{\rho a}.$$

Then $\delta g_{\mu\nu} = h_{\mu\nu}$. Hence the differential is surjective and has rank ten. The domain has dimension sixteen, leaving a six-dimensional kernel. The transformations $\delta e = e\lambda$ with $\lambda^{\mathsf{T}}\eta + \eta\lambda = 0$ lie in the kernel and constitute the six-dimensional Lorentz Lie algebra. □

Corollary 6.2. *The local comparison "a biquaternion has eight real components, whereas a metric has ten" is not the relevant rank count. The metric depends on four covariant first derivatives, which form a sixteen-component real tetrad before local Lorentz redundancy is removed.*

This theorem closes the local algebraic rank question. It does not prove that all tetrad variations arise from admissible variations of one on-shell Θ and its derived connection.

7 Connection and Christoffel symbols

The Christoffel symbols and the frame connection describe different indices:

- $\Gamma^\rho{}_{\mu\nu}$ transports coordinate indices;
- $\omega_\mu{}^a{}_b$, and its spin/biquaternionic lift Ω_μ, transport the local Lorentz frame.

They are related by

$$\partial_\mu e_\nu{}^a - \Gamma^\rho{}_{\mu\nu} e_\rho{}^a + \omega_\mu{}^a{}_b e_\nu{}^b = 0. \tag{7}$$

It is generally incorrect to identify Γ with a real component of Ω.

Theorem 7.1 (Unique torsion-free classical frame connection)**.** *For every nondegenerate C^2 tetrad, metric compatibility and vanishing torsion determine a unique Lorentz connection,*

$$\boxed{\mathring{\omega}_\mu{}^a{}_b = e_b{}^\nu \left(\mathring{\Gamma}^\rho{}_{\mu\nu}(g) e_\rho{}^a - \partial_\mu e_\nu{}^a \right),} \tag{8}$$

with $\mathring{\omega}_{\mu ab} = -\mathring{\omega}_{\mu ba}$. Its spin lift is unique in the chosen representation, up to ordinary local Lorentz gauge.

Proof. The Levi–Civita connection $\mathring{\Gamma}(g)$ is uniquely fixed by $\mathring{\nabla} g = 0$ and $\mathring{\Gamma}^\rho{}_{[\mu\nu]} = 0$. Solving Eq. (7) gives Eq. (8). If two Lorentz connections have the same zero torsion, their difference is antisymmetric in its first two frame indices and symmetric in the indices forced by equality of torsion; cycling these symmetries gives zero. □

The torsion-free result extends to arbitrary specified torsion. Write

$$T^a = de^a + \omega^a{}_b \wedge e^b = \frac{1}{2} T^a{}_{bc} e^b \wedge e^c. \tag{9}$$

Theorem 7.2 (Tetrad–torsion reconstruction)**.** *For every nondegenerate tetrad and specified torsion, there is exactly one metric-compatible Lorentz connection,*

$$\boxed{\omega^a{}_b = \mathring{\omega}^a{}_b(e) + K^a{}_b(T),} \tag{10}$$

where, with the convention above,

$$\boxed{K_{abc} = \frac{1}{2}\left(T_{cab} - T_{abc} - T_{bca}\right).} \tag{11}$$

Proof. Metric compatibility makes K_{abc} antisymmetric in a, b. Since the Levi–Civita part has zero torsion, $T_{abc} = K_{acb} - K_{abc}$. Combining this equation with its cyclic permutations yields Eq. (11), proving existence and uniqueness. □

Thus Ω_μ is not an additional arbitrary classical matter field once the tetrad and torsion are fixed. The full-UBT problem is to derive T from the action. In the GR vacuum branch $T = 0$, so $K = 0$. Because every metric-compatible ω_μ lies in $\mathfrak{so}(1,3)$, its parallel transport preserves η_{ab} and the Lorentz slice, including in branches with nonzero contorsion.

The detailed proof and exact symbolic checks are in `canonical/gr_closure/gap_10omega_connecti` and `tools/verify_gap_10omega_connection.py`.

8 Curvature and special relativity

For a one-sided representation, the curvature is

$$\mathcal{R}_{\mu\nu} = \partial_\mu \Omega_\nu - \partial_\nu \Omega_\mu + [\Omega_\mu, \Omega_\nu]. \tag{12}$$

It must be distinguished from the algebraic tetrad bivector $\Sigma_{\mu\nu}$.

In a flat simply connected region, zero curvature permits a local frame with $\Omega_\mu = 0$. In inertial Cartesian coordinates one can also set $\Gamma^\rho{}_{\mu\nu} = 0$. The flat limit has an explicit single-field representer.

Theorem 8.1 (Affine representer of a constant Lorentz tetrad)**.** *Let $\bar{E}_\mu \in W_L$ be constant and linearly independent. Then*

$$\boxed{\Theta_{\text{aff}}(x) = \Theta_0 + \sqrt{\mathcal{N}_0}\, \bar{E}_\mu x^\mu} \tag{13}$$

in the inertial gauge satisfies $\mathcal{N}_0^{-1/2} \partial_\mu \Theta_{\text{aff}} = \bar{E}_\mu$ and induces the constant metric determined by the central anticommutator.

For

$$\bar{E}_0 = i\mathbf{1}, \qquad \bar{E}_k = \mathbf{e}_k, \tag{14}$$

Eq. (13) becomes

$$\boxed{\Theta_{\text{SR}} = \Theta_0 + \sqrt{\mathcal{N}_0}\left(ix^0\mathbf{1} + x^k \mathbf{e}_k\right),} \tag{15}$$

which generates $g_{\mu\nu} = \eta_{\mu\nu}$. Every second spacetime derivative vanishes, so the standard flat source-free constant-coefficient second-order master operator annihilates Θ_{SR}. Nonzero connection coefficients in curvilinear coordinates do not by themselves imply curvature.

9 Integrability and representation selection

The tetrad is not independent in UBT; it must satisfy $E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta$. Once the classical connection is reconstructed from E and torsion, this becomes an implicit nonlinear first-order system in which the unknown tetrad occurs on both sides.

9.1 One-sided flatness obstruction

Suppose

$$D_\mu^L \Theta = \partial_\mu \Theta + A_\mu \Theta, \qquad [D_\mu^L, D_\nu^L]\Theta = F_{\mu\nu}^L \Theta. \tag{16}$$

Theorem 9.1 (One-sided invertible-field no-go)**.** *Assume $E_\nu = \mathcal{N}_0^{-1/2} D_\nu^L \Theta$, torsion-free antisymmetric tetrad compatibility, the same induced one-sided connection on $D_\nu \Theta$, and invertible Θ. Then $F_{\mu\nu}^L = 0$.*

Proof. Torsion-free compatibility makes the antisymmetrised covariant Hessian vanish, so $F_{\mu\nu}^L \Theta = 0$. Right multiplication by Θ^{-1} gives $F_{\mu\nu}^L = 0$. □

Thus the naive one-sided regular representation cannot be the generic curved GR route for invertible Θ. A noninvertible stabilizer sector is a possible exceptional branch, not a representation of arbitrary geometries.

9.2 Two-sided biquaternionic derivative

The algebra $\mathbb{B}$ is naturally a bimodule over itself. Consider

$$\boxed{D_\mu \Theta = \partial_\mu \Theta + A_\mu \Theta - \Theta B_\mu.} \tag{17}$$

A direct expansion gives

$$\boxed{[D_\mu, D_\nu]\Theta = F_{\mu\nu}^A \Theta - \Theta F_{\mu\nu}^B.} \tag{18}$$

Torsion-free antisymmetric compatibility therefore reduces to

$$F_{\mu\nu}^A \Theta = \Theta F_{\mu\nu}^B. \tag{19}$$

For invertible Θ,

$$\boxed{F_{\mu\nu}^A = \Theta F_{\mu\nu}^B \Theta^{-1}.} \tag{20}$$

The left and right curvatures may be nonzero. The field Θ acts as an intertwiner between them. This does not prove existence for an arbitrary curved tetrad, nor does it prove that the two-sided route is unique; it exhibits a minimal algebra-native escape from the stated one-sided flatness obstruction.

The resulting equation is schematically

$$E_\mu = \mathcal{N}_0^{-1/2} \left[\partial_\mu \Theta + A_\mu[E,T]\Theta - \Theta B_\mu[E,T]\right]. \tag{21}$$

It is an implicit nonlinear PDE/fixed-point system. If Θ is restricted to a Jacobi-theta function class, the complete system may additionally be transcendental; implicitness and transcendental functional dependence are separate mathematical properties.

The full proof and verifier are in `canonical/gr_closure/gap_10i_integrability_selection.tex` and `tools/verify_gap_10i_integrability.py`.

10 Complex time and regularity

The field may depend on $\tau = t + i\psi$. The metric definition Eq. (4) is pointwise; it does not integrate over ψ. A classical four-dimensional sector may require

$$\partial_\psi g_{\mu\nu} = 0,$$

but this must follow from the dynamics or a clearly stated classical-sector condition.

Smoothness of the physical spacetime field and intrinsic biquaternionic regularity in the argument q are different properties. The latter has not yet been uniquely fixed: Cauchy–Fueter, slice-regular, and UBT-specific Jacobi structures must be compared before a regularity axiom is locked.

11 Relation to earlier compact-fiber closure

Earlier work defined a metric by averaging a quadratic form over the compact imaginary-time coordinate and used the resulting function space to repair a rank count. Those constructions remain mathematically auditable, but they are not required by the central anticommutator theorem and are no longer the canonical local metric definition. They are retained in the repository as an exploratory candidate-completion branch and are documented in `canonical/gr_closure/HISTORICAL_FIBER_ROUTE_STATUS.md`. The branch is rejected as canonical because of weak selection and large representer redundancy, not because its mathematics was disproved. In particular, the earlier fiber-average constructions are not used as a canonical derivation of the metric, rank closure, connection, or Einstein dynamics.

The conceptual distinction is:

- the old route enlarged the effective target by retaining a whole ψ profile;
- the present route uses the ordinary tetrad content of the covariant first jet and obtains $16 - 6 = 10$ locally.

12 Schwarzschild and perturbations: honest current status

The repository contains an explicit and numerically verified construction of the spatial isotropic Schwarzschild metric. The static vacuum Einstein relations determine the lapse uniquely once the spatial branch and boundary conditions are assumed. What remains open is the selection of the complete Schwarzschild tetrad by the canonical UBT master dynamics and connection.

The relation $\partial_\psi \Theta = i\Phi(r)\Theta$ is not used as a canonical derivation of the lapse, and a generic Maxwell or $U(1)_\psi$ field does not generate vacuum Schwarzschild. The perturbation bridge must be rederived in the covariant connection/tetrad formulation; earlier fiber-linearisation results remain conditional results of the exploratory branch.

13 Revised gap ledger

Gap	Status
GAP-10K	**CLOSED locally**: tetrad-to-metric differential has rank ten.
GAP-10Ω-KIN	**CLOSED**: specified tetrad and torsion uniquely determine the metric-compatible frame connection.
GAP-10Ω-GR	**CLOSED**: the torsion-free branch is the unique Levi–Civita spin connection, up to Lorentz gauge.
GAP-10T-DYN	**OPEN**: derive vacuum or spin-sourced torsion from the canonical UBT action.
GAP-10L-CONN	**CLOSED**: metric-compatible Lorentz transport preserves η and the Lorentz slice.
GAP-10L-DYN	**OPEN**: prove preservation by the full Θ dynamics and sources.
GAP-10I-SR	**CLOSED**: explicit affine Θ representer for every constant Lorentz tetrad.
GAP-10I-1S	**CLOSED AS NO-GO**: naive one-sided invertible torsion-free branch forces zero curvature.
GAP-10I-2S	**NARROWED**: two-sided derivative avoids the flatness obstruction and yields the curvature-intertwiner condition.
GAP-10I-CURVED	**OPEN**: curved-space existence, uniqueness, regularity, and global continuation.
GAP-10D	**OPEN**: derive Einstein dynamics from the canonical UBT action/master equation.
GAP-10ψ	**OPEN**: classical stability/independence along imaginary time.
GAP-B-MASTER	**OPEN**: perturbation bridge from the original master dynamics.
GAP-U2Θ	**OPEN**: canonical dynamical generation of the full Schwarzschild tetrad/lapse.

14 Conclusion

The clean local UBT geometry is

$$\boxed{\Theta \;\longrightarrow\; E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0} \;\longrightarrow\; \begin{cases} \frac{1}{2}\{E_\mu, E_\nu\}_\sharp = g_{\mu\nu}\mathbf{1}, \\ \frac{1}{2}[E_\mu, E_\nu]_\sharp = \Sigma_{\mu\nu}, \\ [D_\mu, D_\nu]\Theta = F^A_{\mu\nu}\Theta - \Theta F^B_{\mu\nu} \quad \text{in the viable two-sided branch.} \end{cases}}$$

This structure resolves the local rank objection without auxiliary maps, metric projections, or fiber averaging. The connection is also no longer an arbitrary addition: for specified tetrad and torsion it is uniquely reconstructed, and the torsion-free GR branch is Levi–Civita. Special relativity has an explicit affine single-Θ representer. The naive one-sided curved route is ruled out for generic invertible Θ, while the two-sided biquaternionic derivative supplies a nonflat integrability condition. The decisive remaining work is dynamical: derive torsion and the paired left/right connection from the single UBT action, solve the implicit curved system, and obtain Einstein dynamics and observed geometries on shell.

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